Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making

> #EmbodiedAI #Benchmark

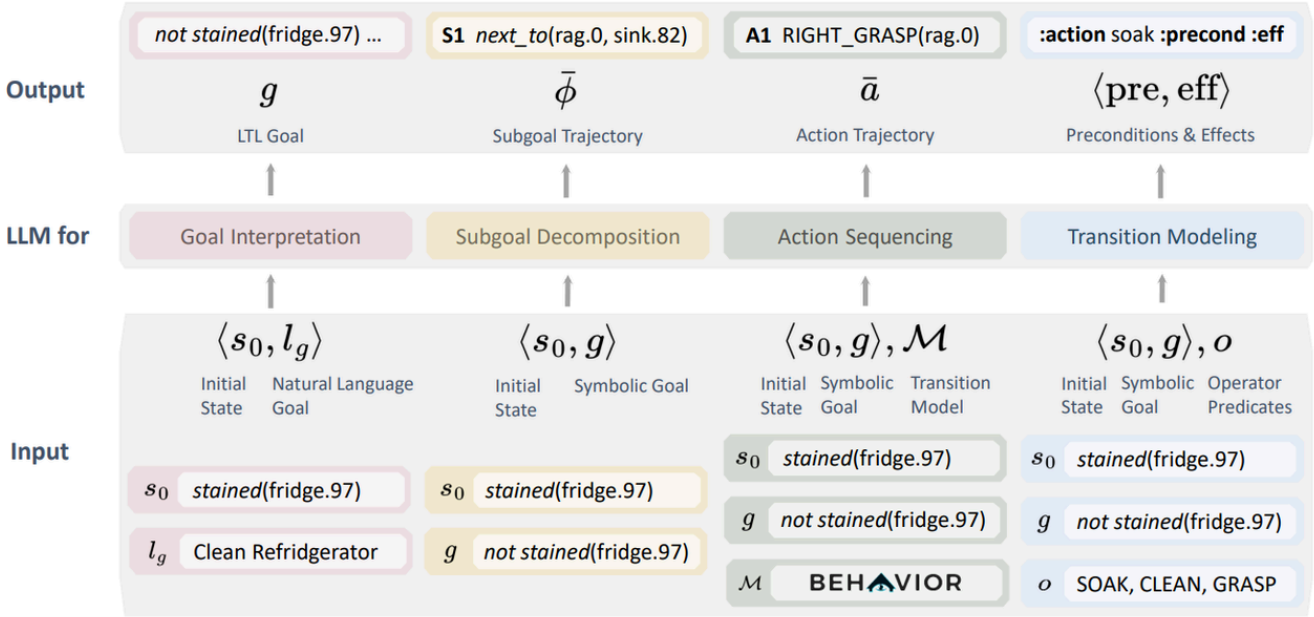

Figure 2: The input and output formulation of four ability modules.

- What problems does this framework solve?
  - Current LLMs have potential to be used in embodied AI, but there lacks benchmarks to evaluate the embodied performance of LLMs.
  - No universal standards; heavily relies on framework.
  - Only focus on the single result (final result), more fine grained benchmarks needed.
- What does this framework do?
  - LTL representations of environment
    - Define the environment as $s = \langle \mathcal{U}, \mathcal{F} \rangle$, where $\mathcal{U}$ is the list of objects and $\mathcal{F}$ is the relational Boolean features (like `isOn(cup, table), isOpen(door)`)
    - To produce universal representations of different simulators, the paper proposes an LTL representation to formally describe the goal, subgoal, action sequencing, and transition modeling.
  - Decoupling of LLM ability:
    1. Goal Interpretation: Receive the instruction from humans and convert it to LTL representations.
    2. Subgoal Decomposition: The LLM needs to breakdown the goal and list a sequence of subgoals' LTL representation.
    3. Action Sequencing: Based on the environmental transition model, generates a sequence of executable actions.
    4. Transition Modeling: Predict the effect and prediction of an action with PDDL format.

- Evaluation Pipeline
  - Isolation: When evaluating one section/ability, the framework will replace other parts with the groundtruth and deterministic parts
  - The framework expands the evaluation from the rate of success to trajectory evaluation+goal evaluation
  - Pinpoint fine-grained errors such as grammar mistakes, hallucinations, planning mistakes, and so on.
- Experiment Results
  - Top-performing models: o1-preview, Claude-3.5 Sonnet, and Gemini 1.5 Pro demonstrated the strongest overall decision-making capabilities. In particular, o1-preview significantly outperformed others in task success rates within the BEHAVIOR environment, which presents long-term planning challenges, while Gemini 1.5 Pro achieved the highest overall accuracy in target interpretation.
  - Weak “Grounding” Capabilities and Lack of Common Sense: LLMs struggle to faithfully map natural language instructions to specific objects and physical states in the environment, often mistaking intermediate states for final goals. Additionally, they frequently overlook the physical common sense implied in everyday human language (for example, in the instruction “put the turkey on the table,” an LLM might ignore the implicit spatial constraint that it must first be placed on a plate).
  - Common runtime errors: When sequencing and planning actions, the most frequent errors made by LLMs are missing steps (often due to overlooking strict prerequisites for actions, such as forgetting to plan the action of opening a door before picking up an object) and redundant steps (continuing to perform ineffective operations even after the goal has been achieved).
  - Relational goals are more challenging than state goals: Compared to altering the state of a single object (such as flipping an appliance switch), LLMs exhibit significantly lower planning success rates and logical formalization accuracy when reasoning about and executing “spatial relationships” or “non-spatial relationships” between objects (such as placing an object on top of another or holding a specific item).
- Work Cited
  - Paper
  - Chat History
  - NotebookLM